On Combiners for Verifiable Random Functions

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Research Question

Can we *combine* multiple instances of verifiable random function (VRFs) into a single VRF that is secure as long as *enough* of the instances are?

- How do we *combine* the candidate schemes?

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Can we combine *k-out-of-n* VRF instances in a *black-box* fashion?

Verifiable Functions

Definition (Verifiable Function)

A *verifiable function* (VF) VF is a triple of the following efficient algorithms:

- $\mathsf{Gen}_{\mathsf{VF}}(1^\lambda) = (\mathsf{sk}, \mathsf{pk})$ generates a key-pair
- $\mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x) = (y, \pi)$ evaluates points to an image-proof pair
- $\mathsf{Vfy}_{\mathsf{VF}}(\mathsf{pk}, x, y, \pi) = b$ verifies evaluations

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A VF is a purely syntactic object, we define security properties on top of it.

Properties of Verifiable Functions

ε-Correctness

Honestly generated image-proof pairs verify with probability at least ε:

$$\Pr\left[\begin{array}{l}(\mathsf{sk},\mathsf{pk}) \leftarrow_{\$} \mathsf{Gen}_{\mathsf{VF}}(1^{\lambda}) \\ \quad x \leftarrow_{\$} \{0,1\}^{\lambda} \quad : \mathsf{Vfy}_{\mathsf{VF}}(\mathsf{pk},x,y,\pi) = 1 \\ (y,\pi) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk},x)\end{array}\right] \geq \varepsilon(\lambda)$$

VF is *correct* if it is 1-correct and *statistically correct* if it is $(1-\varepsilon)$-correct and $\varepsilon \in \mathsf{negl}(\lambda)$.

Properties of Verifiable Functions

Unique Provability

For every public key pk and every point x, at most one image y is provable:

$$\mathrm{Vfy}_{\mathrm{VRF}}(\mathsf{pk}, x, y_0, \pi_0) = \mathrm{Vfy}_{\mathrm{VRF}}(\mathsf{pk}, x, y_1, \pi_1) = 1 \implies y_0 = y_1.$$

Properties of Verifiable Functions

(ε, t)-Pseudorandomness

For any adversary $\mathcal{A}$ running in time at most t:

$$\mathbf{Adv}^{\mathrm{prf}}_{\mathrm{VF}}(\lambda, \mathcal{A}) := 2 \cdot \left| \Pr\left[\mathbf{Exp}^{\mathrm{prf}}_{\mathrm{VF}}(\lambda, \mathcal{A}) = 1\right] - \frac{1}{2} \right| \leq \varepsilon(\lambda)$$

Pseudorandomness experiment

Experiment $\mathbf{Exp}^{\mathsf{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A})$

1 : $b \leftarrow_{\$} \{0,1\}$
2 : $(\mathsf{sk}, \mathsf{pk}) \leftarrow_{\$} \mathsf{Gen}_{\mathsf{VF}}(1^{\lambda})$
3 : $S \leftarrow \varnothing$
4 : $(x^*, \sigma) \leftarrow_{\$} \mathcal{A}_1^{\mathcal{O}_{\mathsf{eval}}}(1^{\lambda}, \mathsf{pk})$
5 : $(y_0, \pi_0) \leftarrow \mathsf{Eval}_{\mathsf{VF}}(\mathsf{sk}, x^*)$
6 : $y_1 \leftarrow_{\$} \mathcal{Y}$
7 : $b' \leftarrow_{\$} \mathcal{A}_2(1^{\lambda}, x^*, \sigma, y_b)$
8 : **return** $b = b' \wedge x^* \notin S$

Oracle $\mathcal{O}_{\mathsf{eval}}(x)$

1 : $S \leftarrow S \cup \{x\}$
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3 : **return** y

Properties of Verifiable Functions

(ε, t)-Unpredictability

For any adversary $\mathcal{A}$ running in time at most t, playing the prf experiment $\mathbf{Exp}^{\mathsf{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A})$:
The advantage $\mathbf{Adv}^{\mathsf{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A})$ is bounded from above by

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Unpredictability experiment

Experiment $\mathbf{Exp}^{\mathrm{prf}}_{\mathrm{VF}}(\lambda, \mathcal{A})$

1 : $b \leftarrow_{\$} \{0,1\}$
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Properties of Verifiable Functions

$(\varepsilon, \mathfrak{t})$-Weak Pseudorandomness

For any adversary $\mathcal{A}$ running in time at most $\mathfrak{t}$, playing the prf experiment $\mathbf{Exp}^{\mathsf{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A})$, the advantage $\mathbf{Adv}^{\mathsf{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A})$ is bounded from above by:

$$\mathbf{Adv}^{\mathsf{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A}) := 2 \cdot \left| \Pr\left[\mathbf{Exp}^{\mathsf{prf}}_{\mathsf{VF}}(\lambda, \mathcal{A}) = 1\right] - \frac{1}{2} \right| \leq \varepsilon(\lambda)$$

A VF is *pseudorandom* if it is $(\varepsilon, \mathfrak{t})$-pseudorandom for $\varepsilon \in \mathsf{negl}(\lambda)$ and $\mathfrak{t} \in \mathsf{poly}(\lambda)$.

Weak Pseudorandomness experiment

Experiment $\mathbf{Exp}^{\mathrm{prf}}_{\mathrm{VF}}(\lambda, \mathcal{A})$

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Verifiable Random Functions

Verifiable Random Function

A *verifiable random function* (VRF) is a VF that is:

- Correct
- Uniquely Provable
- Pseudorandom

Verifiable Unpredictable Functions

Verifiable Unpredictable Function

A *verifiable unpredictable function* (VUF) is a VF that is:

- Correct
- Uniquely Provable
- Unpredictable

Weak Verifiable Unpredictable Functions

Weak Verifiable Unpredictable Function

A *weak verifiable unpredictable function* (WVUF) is a VF that is:

- Correct
- Uniquely Provable
- Weakly Pseudorandom

Why Should We Care About VRFs?

- Signatures with Strong Verifiability Guarantees [GO93]

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- Distributed Leader Election in PoS blockchains [KRDO17; DGKR18; BASV23]

Robust Black-Box Combiners

Definition (Robust $(n; k)$ Black-Box Combiner (Informal))

Robust $(n; k)$ black-box P-combiner Π_P satisfy:

- Give oracle access to n candidate schemes $\mathcal{O}_1, \ldots, \mathcal{O}_n$
- Implements primitive P
- Is secure if at least k candidates are secure

With inefficient we mean adversaries with access to a PSPACE complete oracle.

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- Give oracle access to n candidate schemes $\mathcal{O}_1, \ldots, \mathcal{O}_n$
- Implements primitive P
- Is secure if at least k candidates are secure **even against inefficient adversaries**

With inefficient we mean adversaries with access to a PSPACE complete oracle.

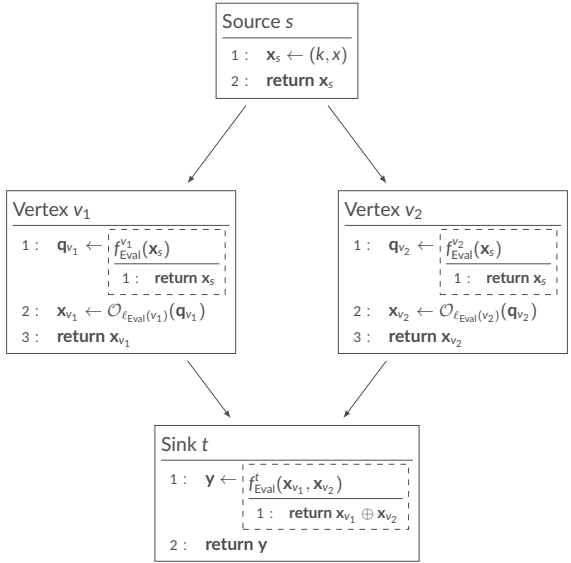
Robust Black-Box DAG Combiners

We consider robust black-box combiners in a DAG normal form.

- Dedicated **source** s (the input) and **sink** t (the output).
- Every other vertex v carries
  - a label $\ell(v) = (i, j)$ specifying *which candidate's* j-th oracle is queried, and
  - a ppt. Turing machine f^v that computes the query input from the responses of the predecessors $\mathcal{N}_{\mathcal{G}}^{-}(v)$.
- Execution traces the DAG in topological order: at each v, build $\mathbf{q}_v \leftarrow f^v(\{\mathbf{x}_w\}_{w \in \mathcal{N}_{\mathcal{G}}^{-}(v)})$ and set $\mathbf{x}_v \leftarrow \mathcal{O}_{\ell(v)}(\mathbf{q}_v)$. The sink's Turing machine f^t produces the final output.

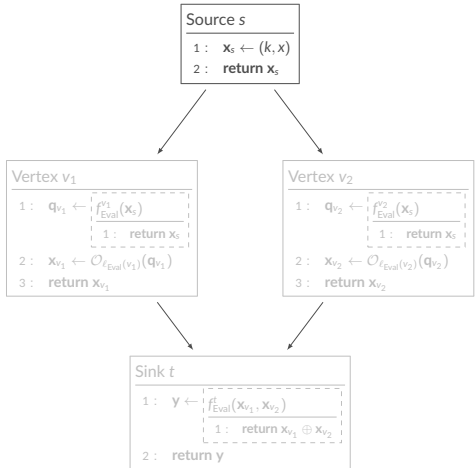
Equivalence. Every robust $(n; k)$ black-box combiner has a *computationally equivalent* DAG combiner — the DAG form is a convenient normal form for our proofs.

Example


Source s
1 : $\mathbf{x}_s \leftarrow (k, x)$
2 : return $\mathbf{x}_s$
Vertex v_1
1 : $\mathbf{q}_{v_1} \leftarrow f^{v_1}_{\text{Eval}}(\mathbf{x}_s)$
1 : return $\mathbf{x}_s$
2 : $\mathbf{x}_{v_1} \leftarrow \mathcal{O}_{\ell_{\text{Eval}}(v_1)}(\mathbf{q}_{v_1})$
3 : return $\mathbf{x}_{v_1}$
Vertex v_2
1 : $\mathbf{q}_{v_2} \leftarrow f^{v_2}_{\text{Eval}}(\mathbf{x}_s)$
1 : return $\mathbf{x}_s$
2 : $\mathbf{x}_{v_2} \leftarrow \mathcal{O}_{\ell_{\text{Eval}}(v_2)}(\mathbf{q}_{v_2})$
3 : return $\mathbf{x}_{v_2}$
Sink t
1 : $\mathbf{y} \leftarrow f^{t}_{\text{Eval}}(\mathbf{x}_{v_1}, \mathbf{x}_{v_2})$
1 : return $\mathbf{x}_{v_1} \oplus \mathbf{x}_{v_2}$
2 : return $\mathbf{y}$

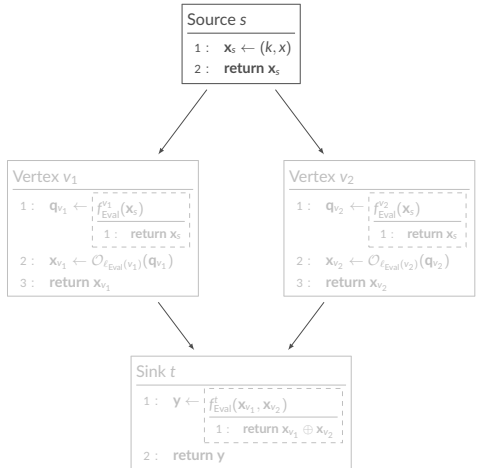
Example

- symbolic, simplifies reasoning
- produces the input



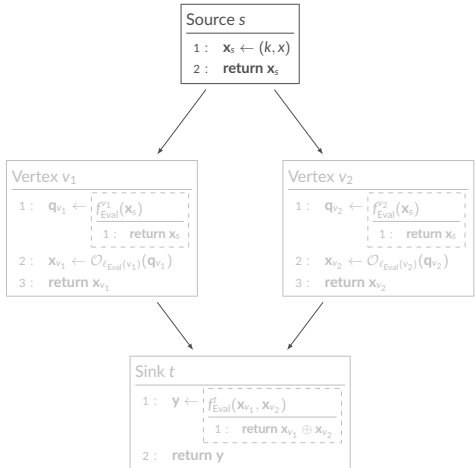
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- Robust $(n; k)$ black-box DAG Combiner $\implies$ Robust $(n; k)$ black-box combiners

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Sufficient to show (im)possibility of a robust $(n; k)$ black-box DAG combiner.

The 1-out-of-2 Combiner

- Simplest non-trivial case, only two candidates

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- Simplest non-trivial case, only two candidates
- Construction extends to robust 1-out-of-*n* black-box combiner
- Impossibility extends to robust $\lfloor n/2 \rfloor$-out-of-*n* black-box combiner

Proof Setup

Suppose Π_{VRF} is a robust $(2;1)$ black-box DAG VRF-combiner.

- $\mathcal{O}_{vrf}$: a secure VRF
- $\mathcal{O}_{uppc}$: a *trivial*, uniquely provable and perfectly correct VF:
  - $\mathsf{Gen}_{\mathsf{VRF}}(1^\lambda) = (\mathsf{sk}_{def}, \mathsf{pk}_{def})$
  - $\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x) = (y_{def}, \pi_{def})$
  - $\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y, \pi) = 1 \iff y = y_{def}$
- $\mathcal{O}_{vrf}$ a secure VRF $\implies$ $\Pi_{\mathsf{VRF}}^{\mathcal{O}_{\mathsf{vrf}}, \mathcal{O}_{uppc}}$ a secure VRF

Goal: Adversary $\mathcal{A}^{\mathsf{PSPACE}, \mathcal{O}_{\mathsf{vrf}}, \mathcal{O}_{uppc}}$ against psuedorandomness of $\Pi_{\mathsf{VRF}}^{\mathcal{O}_{\mathsf{vrf}}, \mathcal{O}_{uppc}}$.
Assume: $\mathcal{SK} = \{0,1\}^{s(\lambda)}$ for $s \in \mathsf{poly}(\lambda)$ for all candidates.

Proof

Assume we have a Turing machine (inefficient) $\mathsf{EvalSim}_{\mathsf{VRF}}$ satisfying:

- $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}_{cand}, x, y, \pi, \tau) = \top$ if $\mathsf{sk}_{cand} = \mathsf{sk}$
- $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}_{cand}, x, y, \pi, \tau) = \bot$ if $\mathsf{Eval}^{y}_{\mathsf{VRF}}(\mathsf{sk}_{cand}, x) \neq \mathsf{Eval}^{y}_{\mathsf{VRF}}(\mathsf{sk}, x)$
- $\mathsf{EvalSim}_{\mathsf{VRF}}$ makes *no queries to any oracles*.
- $\mathsf{EvalSim}_{\mathsf{VRF}}$ has a *polynomial space complexity*.

For $x \in \{0,1\}^{\lambda}$, $(y, \pi) = \Pi^{\mathcal{O}_{vrf}, \mathcal{O}_{uppc}}_{\mathsf{VRF}} \mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x)$ and $\tau = \mathsf{trace}(\Pi^{\mathcal{O}_{vrf}, \mathcal{O}_{uppc}}_{\mathsf{VRF}}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y, \pi))$.

Adversary

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Adversary

$$\mathbf{Adv}^{\mathsf{prf}}_{\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf},\mathcal{O}_{uppc}}}(\lambda,\mathcal{A}) \geq \frac{1}{2} - \frac{1}{2^{\lambda+1}} - \frac{1}{|\mathcal{Y}|}.$$

How to get $\mathsf{EvalSim}_{\mathsf{VRF}}$

Given $(\mathsf{sk}_{cand}, x, y, \pi, \tau)$, $\mathsf{EvalSim}_{\mathsf{VRF}}$ is computed by:

- Follow the DAG $\mathcal{G}_{\mathsf{Eval}}$, describing $\Pi_{\mathsf{VRF}}^{\mathcal{O}_{vrf}, \mathcal{O}_{uppc}}$, in some topological ordering
- For every vertex v corresponding to a query to $\mathcal{O}_{uppc}$ answer with default value
- For every vertex v corresponding to a query to $\mathcal{O}_{vrf}$ check τ:
  - If $\mathbf{q} \in \tau$: answer accordingly
  - If $\mathbf{q} \notin \tau$: branch for every possible answer
- Return $\top$ iff all branches produce the image y

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- $\mathsf{EvalSim}_{\mathsf{VRF}}$ has a *polynomial space complexity*.

For $x \in \{0,1\}^{\lambda}, (y, \pi) = \Pi^{\mathcal{O}_{vrf}, \mathcal{O}_{uppc}}_{\mathsf{VRF}} \mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}, x)$ and $\tau = \mathsf{trace}(\Pi^{\mathcal{O}_{vrf}, \mathcal{O}_{uppc}}_{\mathsf{VRF}}.\mathsf{Vfy}_{\mathsf{VRF}}(\mathsf{pk}, x, y, \pi))$.

How to get $\mathsf{EvalSim}_{\mathsf{VRF}}$

$\mathsf{EvalSim}_{\mathsf{VRF}}$ has a *polynomial space complexity*:

- $\mathsf{EvalSim}_{\mathsf{VRF}}$ in NPSPACE since every branch is an execution of $\mathsf{Eval}_{\mathsf{VRF}}$
- NPSPACE $\subseteq$ PSPACE

How to get $\mathsf{EvalSim}_{\mathsf{VRF}}$

$\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}_{cand}, x, y, \pi, \tau) = \perp$ if $\mathsf{Eval}^{y}_{\mathsf{VRF}}(\mathsf{sk}_{cand}, x) \neq y$:

- All queries made to $\mathcal{O}_{uppc}$ answered the same $\mathsf{Eval}_{\mathsf{VRF}}(\mathsf{sk}_{cand}, x)$
- All queries made to $\mathcal{O}_{vrf}$ with $\mathbf{q} \in \tau$ answered the same
- For all queries made to $\mathcal{O}_{vrf}$ with $\mathbf{q} \notin \tau$ there exists a branch which is the same

$\implies$ At least one branch evaluates to $\mathsf{Eval}^{y}_{\mathsf{VRF}}(\mathsf{sk}_{cand}, x) \neq y$.

How to get $\mathsf{EvalSim}_{\mathsf{VRF}}$

$\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}_{cand}, x, y, \pi, \tau) = \top$ if $\mathsf{sk}_{cand} = \mathsf{sk}$.

- Suppose $\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}, x, y, \pi, \tau) = \bot$
- Then there exist answers to queries to $\mathcal{O}_{vrf}$ such that the output is $y' \neq y$
- Modify $\mathcal{O}_{uppc}$ to make it a secure VRF such that $\mathcal{O}^*_{uppc}|_{\widetilde{\mathcal{SK}}} = \mathcal{O}_{uppc}|_{\widetilde{\mathcal{SK}}}$
- Modify $\mathcal{O}_{vrf}$ to produce the transcript of the branch, $\mathcal{O}^*_{vrf}$
- $\mathcal{O}_{vrf}$ and $\mathcal{O}^*_{vrf}$ only differ on queries $\mathbf{q} \notin \tau$ and indistinguishable for verification

Now we have:

$$y = \Pi_{\mathsf{VRF}}^{\mathcal{O}^*_{uppc}, \mathcal{O}_{vrf}}.\mathsf{Eval}^y_{\mathsf{VRF}}(\mathsf{sk}, x) \neq \Pi_{\mathsf{VRF}}^{\mathcal{O}^*_{uppc}, \mathcal{O}^*_{vrf}}.\mathsf{Eval}^y_{\mathsf{VRF}}(\mathsf{sk}, x) = y^*$$

How to get $\mathsf{EvalSim}_{\mathsf{VRF}}$

$\mathsf{EvalSim}_{\mathsf{VRF}}(\mathsf{sk}_{cand}, x, y, \pi, \tau) = \top$ if $\mathsf{sk}_{cand} = \mathsf{sk}$.
Now we have:

$$y = \Pi_{\mathsf{VRF}}^{\mathcal{O}^*_{uppc}, \mathcal{O}_{vrf}}.\mathsf{Eval}^{y}_{\mathsf{VRF}}(\mathsf{sk}, x) \neq \Pi_{\mathsf{VRF}}^{\mathcal{O}^*_{uppc}, \mathcal{O}^*_{vrf}}.\mathsf{Eval}^{y}_{\mathsf{VRF}}(\mathsf{sk}, x) = y^*$$

Since the verification algorithms are however the same on queries $\mathbf{q} \in \tau$, the proof π verifies identically in both executions.

$\implies$ Two distinct values $y \neq y^*$ both verify under the same (pk, x, π), breaking **unique provability** of Π_{VRF}.

Construction of $\mathcal{O}^*_{uppc}$

TODO: Move to backup

From $(2; 1)$ to $(n; \lfloor n/2 \rfloor)$

Given a robust $(n; \lfloor n/2 \rfloor)$ black-box VRF-combiner Π_{VRF}, build a robust $(2; 1)$ combiner $\widetilde{\Pi}_{\mathsf{VRF}}$:

- initialize the first $\lfloor n/2 \rfloor$ candidates of Π_{VRF} with $\mathcal{O}_1$
- initialize the remaining $\lceil n/2 \rceil$ candidates with $\mathcal{O}_2$

If either $\mathcal{O}_1$ or $\mathcal{O}_2$ is a secure VRF, at least $\lfloor n/2 \rfloor$ candidates of Π_{VRF} are secure, so $\widetilde{\Pi}_{\mathsf{VRF}}$ is a robust $(2; 1)$ combiner.

Contraposition with the main theorem yields the impossibility of robust $(n; \lfloor n/2 \rfloor)$ black-box VRF-combiners.

Result

Theorem. Impossibility 1-out-of-2 combiner.

There is **no** robust $(2; 1)$ black-box VRF-combiner.

Corollary. Impossibility $\lfloor n/2 \rfloor$-out-of-n combiner.

There is **no** robust $(n; \lfloor n/2 \rfloor)$ black-box VRF-combiner.

Conjecture. Impossibility $(n-1)$-out-of-n combiner.

There is **no** robust $(n; n-1)$ black-box VRF-combiner.

What We Showed

Additionally the same proof shows that:

Lemma. Impossibility $\lfloor n/2 \rfloor$-out-of-n VUF-combiner.

There is **no** robust $(2; 1)$ black-box VRF-combiner.

Lemma. Impossibility $\lfloor n/2 \rfloor$-out-of-n wVRF-combiner.

There is **no** robust $(2; 1)$ black-box VRF-combiner.

Beyond Impossibility

Let assume *unique provability* for all *n* candidates:

- *Then a parallel combiner for PRF is applicable*

Too strong of an assumption?

- No, unique provability is an information theoretic property and typically proven.

Robust 1-out-of-n Black-Box Combiner for UP-VRF

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- Established impossibility for robust $(n; \lfloor n/2 \rfloor)$ black-box VRF-combiner
- Conjectured impossibility for robust $(n; n - 1)$ black-box VRF-combiner

Summary

- Introduced Robust $(n; k)$ Black-Box DAG Combiner for more principled analysis
- Established impossibility for robust $(n; \lfloor n/2 \rfloor)$ black-box VRF-combiner
- Conjectured impossibility for robust $(n; n-1)$ black-box VRF-combiner
- Constructed robust $(n; 1)$ black-box UP-VRF-combiner

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Where the Proof breaks down

- In the 1-out-of-2 case we rely on the fact that for one candidate we can always guess the responses perfectly, and for the other the obscure queries are mutually irrelevant in the sense that under the correct secret key the image y stays the same. That means that for all tests on the correct secret keys sk^* the output remains consistent through all runs. For any $n > 2$ and $k = n - 1$ we have at least two candidate whose outputs we cannot predict perfectly. While mutual independence holds for